

Title: Preventing dataset shift from breaking ML biomarkers

Aim

The primary aim of the paper is to characterise the problem of dataset shift in the context of machine-learning-extracted biomarkers and to provide strategies for its detection and correction.

Specifically, the authors seek to:

- Explain the risks of dataset shift: The paper highlights how a mismatch between a training cohort and a target population - often caused by recruitment biases - can undermine the reliability of biomarkers when they are applied to new individuals in clinical practice.
- Provide mitigation strategies: It offers an overview of various methodological approaches to make biomarkers more robust, with a particular focus on importance weighting as a tool to create a "pseudo-sample" that reflects the target population.
- Clarify misconceptions: A key goal is to distinguish dataset shift from other problems like confounding, specifically arguing that "deconfounding" (regressing out covariates) is a "false solution" that typically degrades prediction performance rather than fixing shift issues.
- Establish evaluation standards: The paper aims to promote rigorous evaluation through independent test sets and cross-validation schemes that account for heterogeneity, such as holding out entire sites in multi-site datasets.
- Promote fairness and reliability: Ultimately, the research aims to provide recommendations for designing fair and generalisable biomarkers so that machine-learning models can be safely and effectively deployed in healthcare applications.

Theoretical Framework / Key Concepts

The theoretical framework of the research focuses on the intersection of supervised machine learning and biomedical research, specifically addressing how discrepancies between data populations can undermine the reliability of biomarkers.

Core Machine Learning Principles

- Empirical Risk Minimization: This is the central principle used to identify biomarkers. A supervised learning algorithm seeks a function (f) that minimizes the empirical risk - the average error (loss) measured across available "training" examples.
- Loss Function (L): A mathematical tool used to measure the distance between a model's prediction ($f(X)$) and the true value (Y).
- Source vs. Target Data: The source data is the sample used to fit and evaluate a model (often a research cohort), while the target data represents the population where the biomarker is intended for clinical use.
- Overfitting: This occurs when a flexible model learns rules that fit training examples too tightly, causing it to fail when generalizing to new, unseen individuals.
- External Validity: The process of assessing whether a predictor maintains its performance outside of the specific dataset used to define it.

Key Concepts of Dataset Shift

Dataset shift (or dataset bias) refers to a mismatch between the distribution of the training cohort and the target population.

- Covariate Shift: A scenario where the distribution of input features (X) changes between the source and target, but the underlying relationship to the outcome ($P(Y|X)$) remains the same.

- **Prior Probability Shift (Label/Target Shift):** Occurs when the distribution of the output (Y) changes - such as a change in disease prevalence - while the way the condition manifests ($P(X|Y)$) stays constant.
- **Preferential Sample Selection:** A common source of shift that happens when members of a population do not have equal probabilities of being included in the source dataset due to factors like volunteering self-selection or recruitment biases.
- **Spectrum Effects:** When a dataset captures only a small range of possible patient manifestations, leading to systematic errors.

Mitigation and Evaluation Concepts

- **Importance Weighting:** A strategy that reweights available data to create a "pseudo-sample" reflecting the target population's distribution. This is often achieved using inverse probability weights (or propensity scores).
- **"Deconfounding" (Regressing-out):** A procedure from causal analysis that is often misapplied to prediction tasks. The authors define it as a "false solution" because removing the effect of a correlate often destroys the valuable predictive signal.
- **Fairness:** The necessity of ensuring a predictor performs well across all subpopulations (e.g., age, sex, ethnicity) to prevent discriminative clinical decisions for under-represented groups.
- **Domain Adaptation:** A broad category of learning problems and methods (including dataset shift) that aims to make models resilient to changes in data distribution.

Methodological Concepts

The methodological concepts in the research focus on the mathematical principles of machine learning and the specific strategies used to detect and mitigate dataset shift in biomedical biomarkers.

Core Machine Learning Concepts

- **Empirical Risk Minimization (ERM):** This is the foundational process used to identify biomarkers. An algorithm seeks a function (f) that minimizes the empirical risk, which is the average error (measured by a loss function) over a set of available "training" examples.
- **Loss Function (L):** A mathematical tool used to measure the discrepancy between a model's prediction and the actual true value.
- **Training vs. Target Data:** Methodologically, it is crucial to distinguish between source data (used to fit and evaluate the model) and target data (where the model will be applied in clinical practice).

Evaluation Strategies

- **Independent Test Sets and Cross-Validation:** To avoid overfitting, performance must be measured on data held out during training. Cross-validation involves repeating this process with multiple different splits of the data.
- **External Validity:** This involves assessing model performance on a completely independent dataset collected by a different research group or at a different time to ensure the associations found are not unique to the original sample.
- **Site-level Validation:** For datasets collected across multiple hospitals or devices, the authors recommend holding out entire sites during cross-validation and aggregating performance scores at the site level rather than pooling all individuals together.

Mitigation and Correction Strategies

- **Importance Weighting:** A generic approach where training examples are reweighted by the ratio of their likelihood in the target data versus the source data. This creates a "pseudo-sample" that better reflects the population where the biomarker will be used.
- **Inverse Probability Weighting (IPW):** A specific type of weighting used to correct for preferential sample selection. Examples that had a low probability of being selected for the study receive higher weights to account for similar individuals who were not included.

- **Distributionally Robust Optimization (DRO):** Instead of targeting one specific population, this method minimizes the worst-case risk across a set of potential target distributions (an "uncertainty set") to ensure the model is robust to various shifts.
- **Bayes' Rule Adjustment:** A method used specifically to correct prior probability shift (e.g., changes in disease prevalence). It allows researchers to adjust a model's predicted probabilities without needing to extract a new biomarker.
- **Data Augmentation:** The use of generative models or known invariances (like flipping or rotating images) to synthetically increase the diversity of the training set.

False or Misapplied Methodologies

- **"Deconfounding" (Regressing-out):** This involves removing the statistical effect of a variable (like age) from the features before training. The authors argue this is a "false solution" for prediction tasks; while useful in causal analysis, in a machine learning context, it often destroys the informative signal and degrades prediction performance.

Assumptions

The research highlights several critical assumptions that underpin machine learning for biomarkers and the specific strategies used to address dataset shift:

General Machine Learning Assumptions

- **Population Consistency:** The most fundamental assumption in supervised learning (specifically empirical risk minimization) is that the model will be applied to individuals drawn from the same population as the training data. If this assumption is violated, the empirical risk becomes a poor estimate of the model's true clinical performance.
- **Uniform Selection:** Standard validation techniques, such as randomly splitting a dataset for cross-validation, assume that training and test examples are chosen uniformly from the same sample, meaning no dataset shift is present.
- **Predictive Association:** The models assume that the statistical associations captured in the features (X) are relevant to the output (Y), whether these links are causal or not.

Methodological Assumptions for Dataset Shift

- **Support Overlap (Completeness):** For importance weighting to work, the source distribution must cover the same range as the target distribution. It is assumed that no part of the target distribution is completely unseen in the training data; for instance, if women are entirely missing from a study, weighting cannot correct the resulting bias.
- **Selection Probability:** When correcting for preferential sample selection, the methodology assumes that every individual in the target population has a non-zero probability of being selected for the source dataset.
- **Correct Model Specification:** In scenarios of covariate shift, it is assumed that a correctly specified model trained with uniform weights can still yield optimal predictions. However, this assumption fails if the model is over-constrained (misspecified), such as a linear model trying to fit complex non-linear data.
- **Distributional Invariance:** Effective correction strategies often rely on the hypothesis that certain distributions remain unchanged in the target data. For example, in covariate shift, it is assumed that while the input features (X) may change, the conditional distribution of the outcome given the features ($P(Y|X)$) stays the same. In prior probability shift, it is assumed that while disease prevalence ($P(Y)$) changes, the way the condition manifests ($P(X|Y)$) is constant.

Statistical and Mathematical Assumptions

- **Admission of Densities:** The researchers implicitly assume that all random variables analyzed admit probability densities.

- Independence of Auxiliary Data: When computing importance weights, it is assumed that additional information used for prediction that is independent of the features and outcomes should be excluded to avoid increasing the variance of the weights.

Methodology

The methodology mentioned in the sources focuses on the principles of supervised machine learning for biomarker identification, the rigorous evaluation of these models, and specific strategies to mitigate dataset shift.

Core Machine Learning Methodology

- Empirical Risk Minimization (ERM): This is the foundational process for identifying biomarkers. A supervised learning algorithm seeks a function (f) that minimizes the empirical risk ($\hat{R}$), defined as the average error (measured by a loss function L) across a set of available training examples.
- Loss Function Selection: The methodology uses a loss function to measure the distance between a model's "best guess" ($f(X)$) and the true output (Y), such as the squared difference in regression tasks.

Evaluation and Validation Methods

- Independent Test Sets: To obtain valid estimates of performance and avoid overfitting, error is measured on a sample of data that was held out during the training phase.
- Cross-Validation: This involves randomly splitting the available data into training and testing sets multiple times to ensure the results are robust and not unique to a single split.
- External Validation: Models are tested on completely independent datasets collected by different researchers or at different times to assess external validity.
- Site-level Validation: For multi-site datasets, the authors recommend holding out entire sites during cross-validation and aggregating performance scores at the site level rather than pooling all individuals together.

Strategies for Correcting Dataset Shift

- Importance Weighting: A generic tool where training examples are reweighted by the ratio of their likelihood in the target population versus the source population. This creates a "pseudo-sample" that better reflects the clinical population where the biomarker will be applied.
- Inverse Probability Weighting (IPW): A specific type of weighting used to correct for preferential sample selection. Individuals who had a low probability of being included in the original study receive higher weights to represent similar individuals who were not selected.
- Distributionally Robust Optimization (DRO): A more advanced approach that minimizes the worst-case risk across an "uncertainty set" of potential target distributions, ensuring the model performs reliably even if the exact target distribution is unknown.
- Bayes' Rule Adjustment: A method specifically used to handle prior probability shift (e.g., changes in disease prevalence). It allows for the adjustment of a model's predicted probabilities without needing to re-extract the biomarker.
- Invariance Learning: Techniques like domain-adversarial learning that strive to extract features that are predictive of the outcome while remaining invariant to the factors causing the dataset shift.

Experimental Case Study Methodology

The paper illustrates these concepts through a specific experiment predicting tobacco smoking status in the UK Biobank:

- Feature Selection: 29 relevant predictive features were identified using gradient boosting and permutation feature importance on a preliminary set of 6,000 participants.

- **Simulation of Shift:** Two samples of 100,000 individuals were drawn with intentionally different age distributions ("young" vs. "old") to test how training on one population affected performance on the other.
- **Model Comparison:** The researchers compared a linear support vector classifier (over-constrained) against a gradient boosting model (flexible) across three strategies: an unmodified baseline, importance weighting, and regressing-out age.

Methodologies to Avoid

- **Deconfounding (Regressing-out):** The authors explicitly label "deconfounding" correlates of the shift (like age) as a "false solution" for predictive models, as it often destroys the informative signal and degrades prediction performance in all configurations.
- **Homogeneous Sampling:** Selecting participants to be as similar as possible is discouraged for prediction tasks, as it leads to spectrum effects and degrades the model's fairness and ability to generalize.

Main Findings

The research identifies several critical findings regarding how dataset shift impacts machine-learning biomarkers and the effectiveness of various strategies to mitigate these issues.

1. General Impact of Dataset Shift

- **Performance Degradation:** The primary finding is that a mismatch between training (source) and application (target) populations significantly undermines a biomarker's reliability. For instance, in a study predicting tobacco smoking status, training on a younger population and testing on an older one (or vice versa) consistently led to reduced prediction scores.
- **Systematic Error:** Dataset shift causes systematic biases that cannot be corrected simply by increasing the amount of data from the same sources; specific methodological adjustments are required.

2. Failure of "Deconfounding" in Prediction

- **A "False Solution":** One of the paper's most significant findings is that "regressing out" or "deconfounding" covariates associated with the shift (such as age or site indicators) is detrimental to predictive models.
- **Signal Destruction:** While useful in causal analysis, this practice in a machine-learning context often destroys the informative signal necessary for prediction. The authors found that this approach degraded performance in all tested configurations, regardless of whether a shift was present.

3. Model Flexibility and Robustness

- **Flexible Models are More Robust:** Powerful, non-linear models (such as gradient boosting or RBF-SVMs) were found to be more resilient to dataset shift than simpler, over-constrained linear models.
- **Capturing Diverse Patterns:** Flexible models can often capture the correct classification for various subgroups (e.g., both young and old individuals) simultaneously, whereas linear models might only fit a restricted region of the feature space.

4. Utility and Risks of Importance Weighting

- **Benefit for Simple Models:** Reweighting training examples to match the target population (importance weighting) can improve the performance of simpler, misspecified models.
- **Risk for Powerful Models:** Conversely, for flexible learners that already generalize well, importance weighting can be detrimental. It does not provide additional benefit but increases the variance of the empirical risk, effectively reducing the training sample size.

5. Findings on Data Diversity and Multi-site Datasets

- Harm of Homogeneous Sampling: Selecting participants to be as similar as possible (to reduce variance) actually degrades prediction performance and fairness. Large, diverse datasets are essential for creating models that generalize well in clinical settings.
- Multi-site Heterogeneity: In datasets collected across multiple hospitals or devices, the researchers found that site effects mirror dataset shifts. Rather than regressing out site variables, it is more effective to model the variance across sites and perform validation by holding out entire sites during testing.

Limitations

The sources identify several critical limitations, risks, and conditions under which machine-learning biomarkers may fail when applied in clinical practice.

Data Collection and Selection Limitations

- Recruitment and Selection Biases: Preferential sample selection - where certain members of a population (e.g., specific age groups, ethnicities, or socioeconomic statuses) are more likely to be included than others - is ubiquitous and often cannot be fully prevented by study design.
- Homogeneous Sampling Risks: While selecting very homogeneous datasets can reduce variance in statistical testing, it is a limitation in predictive settings because it degrades prediction performance and fairness. This can lead to "spectrum effects" where the model only understands a small range of disease manifestations.
- The "Large Data" Misconception: Systematic dataset biases cannot be fixed simply by acquiring larger datasets from the same sources; they require specific methodological interventions.

Validation and Evaluation Gaps

- Illusory Success of Cross-Validation: Standard cross-validation scores on a single dataset do not guarantee that a model will perform well on clinical data.
- Limits of Independent Test Sets: While testing on an independent dataset is useful for checking external validity, it is "no silver bullet". Recruitment biases (such as difficulty recruiting patients with severe symptoms) may be similar across independently collected datasets, meaning they might share the same underlying shifts.

Methodological Failures and Risks

- The "Deconfounding" Fallacy: Attempting to "deconfound" or "regress out" covariates associated with the shift (like age or site indicators) is a "false solution" in predictive settings. This practice often destroys the informative signal necessary for prediction, degrading performance across all configurations.
- Invariance Learning Risks: Similarly, striving to make features invariant to the effect driving the shift can inadvertently remove valuable signals if those effects are not independent of the medical outcome of interest.
- Limitations of Importance Weighting: While useful, this correction strategy has significant constraints:
 - It cannot fix the shift if a subpopulation is completely missing from the training data (e.g., a model trained only on men cannot be weighted to represent women).
 - It requires specific knowledge of the target distribution and that the source data covers the same "support" (range) as the target.
 - It can increase the variance of the risk estimate, which may actually degrade the performance of powerful, flexible learners like gradient boosting.

Technical and Practical Constraints

- Data Harmonization Difficulties: Compensating for differences across different measurement devices or scanners is notoriously difficult and imperfect. Removing too much inter-site variance can lead to a loss of the very signal the biomarker is trying to detect.

- **Clinical Readiness:** Winning performance in a research competition does not mean a model is ready for the hospital; winning models often achieve accuracy on datasets with strict selection criteria (e.g., excluding patients with comorbidities) that do not reflect the diverse general population.

Implication of Practise

Data Auditing and Diversity

A critical first step is systematically auditing training data for recruitment bias by comparing the demographic and clinical profiles of the training sample against the intended clinical population. Practitioners are advised to prioritise dataset diversity - including demographic, clinical, and geographic variety - over sheer data volume or homogeneity, as diversity more reliably improves generalisation. Additionally, common practices like "regressing out" demographic or site variables should be avoided, as these factors often contain genuine predictive signals that improve performance across populations.

Rigorous Validation Strategies

The sources advocate for a move away from standard random cross-validation, which often yields overly optimistic results. Instead, practitioners should use:

- **Population-Level Holdouts:** Evaluating models by holding out entire sites, cohorts, or time periods.
- **Site-Held-Out Cross-Validation:** This should be the default in multi-site studies to ensure the model is not relying on site-specific artifacts.
- **Critical External Validation:** While independent external validation is necessary, it is not a "final verdict" of readiness, as independent datasets may still share systematic biases with the training data.
- **Prospective Validation:** High performance on benchmarks or research competitions is insufficient; models require prospective validation in the actual intended deployment setting.

Model Selection and Technical Adjustments

The choice of model architecture and adjustment techniques significantly impacts robustness:

- **Model Flexibility:** Flexible non-linear models (e.g., ensemble or kernel-based methods) are preferred for heterogeneous populations because they can learn rules that hold across diverse subgroups simultaneously.
- **Importance Weighting:** This can correct for known sampling biases in simpler or misspecified models but may be counterproductive for highly flexible learners like neural networks.
- **Recalibration:** When disease prevalence in the deployment setting differs from the training data, Bayes' rule adjustment can be used to recalibrate predicted probabilities.
- **Uncertainty Management:** If the target population is unknown, distributionally robust optimisation can be used to minimise worst-case risks across various plausible distributions.

Safety and Continuous Monitoring

Deployment is viewed as an ongoing process rather than a final step. Practitioners must monitor subpopulation performance as a core safety metric to ensure the model does not systematically fail for specific patient groups, such as minority populations or those with rare disease subtypes. Before deployment, simple distribution diagnostics (e.g., summary statistics or visualisations) should be used to detect dataset shift. Finally, deployment pipelines must be designed for continuous monitoring and adaptation, allowing for the detection of performance drift and the triggering of necessary recalibrations throughout the model's lifecycle.